

Dr. Derick Nganyu Tanyu

AI/ML RESEARCH SCIENTIST (PH.D) · SCIENTIFIC COMPUTING (M.Sc.) · TELECOMMUNICATIONS ENGINEER (M.Eng.)

Bad Schönborn, Germany

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Experience

SAP SE

St. Leon-Rot, Germany

DEVELOPMENT SPECIALIST – QUALITY AND CROSS FUNCTIONS

Jan. 2025 - PRESENT

- Develop AI-driven quality and test concepts for SAP S/4HANA Private Cloud.
- Support GenAI initiatives (Joule, AI Core, SEO AI) for enterprise quality engineering.
- Define AI testing strategies, metrics, and quality guidelines.
- Build data-driven quality analyses, reports, and visualizations for stakeholders.
- Bridge ML research and product engineering with practical AI solutions.
- Test plan management for S4/HANA support packages.

ZeTeM (Zentrum für Technomathematik), University of Bremen

Bremen, Germany

POSTDOCTORAL RESEARCH SCIENTIST

Dec. 2023 - Dec. 2024

- Scientific machine learning applied to damage detection in fibre metal laminates.
- Inverse Problems and Parameter Identification.

RESEARCH SCIENTIST

Dec. 2020 - Nov. 2023

- Applying Data-driven methods (Deep Learning) Methods to Partial Differential Equations.
- Parameter Identification Problems in the automotive industry (vehicle development process).
- Inverse Problems and Parameter Identification.
- Time Series forecasting with RNN, LSTM, GRU.

HUAWEI Technologies Co. Ltd

Douala & Yaounde, Cameroon

CONSULTANT– DATACOM (DATA/IP NETWORK)

Jul. 2015 - Aug. 2018

- Configuration, and troubleshooting of IP connectivity and transmission links for ISP Customers.
- Configuration of Datacentre Switches for Server Hosting and Collocation Services at MTNC Datacenters.

CAMTEL (Cameroon Telecommunications)

Yaounde, Cameroon

ENGINEERING INTERN AND RESEARCHER FOR BTS ENERGY OPTIMIZATION

Feb. 2015 - Jul. 2015

- Development of software to forecast energy utilization and thus, provision of optimal functioning (based on Particle Swarm Optimization) taking into consideration environmental friendliness as well as operation and maintenance costs.
- Study the percentage of Energy consumed by existing BTS and Preventive maintenance at remote sites.

Eneo (Energy of Cameroon)

Douala, Cameroon

ENGINEERING INTERN

Aug. 2014 - Oct. 2014

- Study of the SCADA system in Cameroon's National grid for the purpose of monitoring and effective intervention.
- Preventive maintenance at remote sites.

Education

ZeTeM (Zentrum für Industriemathematik), University of Bremen

Bremen, Germany

PHD IN INDUSTRIAL MATHEMATICS

Dec. 2020 - Nov. 2023

- Thesis: *On the Interplay between Deep Learning, Partial Differential Equations and Inverse Problems*. Supervisors: Prof. Dr. Peter Maass. This thesis investigates, with numerical studies, the 'new' field of scientific (physics-informed) machine learning which seeks to solve PDEs with deep learning (physics-informed neural networks and neural operators). Application to a continuum mechanics problem in the automobile industry is done as well as on the Calderón problem (electrical impedance tomography).

University of Ghana & AIMS (African Institute for Mathematical Sciences)

Accra, Ghana

M.SC. IN MATHEMATICAL SCIENCES

Aug. 2018 - Jun. 2019

- Thesis: *Reaction-diffusion Equations with Spatially-varying Parameters.*

Supervisor: Prof. Dr. Georg Bader. A semi-implicit numerical scheme is developed for the reaction-diffusion equation with a space-dependent parameter. Numerical studies and stability analysis are performed. A speed-up is achieved when compared with the explicit schemes.

ENSP(Ecole Nationale Supérieure Polytechnique), University of Yaounde I

Yaounde, Cameroon

M.ENG. IN TELECOMMUNICATIONS ENGINEERING

Sep. 2010 - Jul. 2015

- Thesis: *Optimization of the Hybrid Energy Sources Powering On-grid CAMTEL Base Stations.* Proposed a method based on particle swarm optimization for optimal use of the energy supply at remote base stations, while saving cost.

Publications and Preprints (selection)

- 2023 Nganyu Tanyu D. et al.: *Deep Learning Methods for Partial Differential Equations and Related Parameter Identification Problems*, Inverse Problems
- 2023 Nganyu Tanyu D. et al.: *Parameter Identification by Deep Learning of a Material Model for Granular Media*, arXiv preprint
- 2023 Nganyu Tanyu D. et al.: *Electrical Impedance Tomography: A Fair Comparative Study on Deep Learning and Analytic-based Approaches* arXiv preprint
- 2023 Nganyu Tanyu D.: *From Neural Operators to Complex PDE-based Inverse Problems: Comparative Methods for Problem Solving*, In preparation
- 2022 Kibula L. T., Nganyu Tanyu D. et al.: *Long Term Electricity Load Forecast Based on Machine Learning for Cameroon's Power System*, Energy and Environment Research.

Skills

Languages	English (Mother tongue), French (Mother tongue), German (Working Proficiency)
Programming	Python, Matlab, $\LaTeX$, R (Basics), C++ (Basics)
Deep Learning	Pytorch, TensorFlow, SK-learn.
System Design	Huawei ENSP, Packet Tracer, Matlab Simulink, Proteus.
IP Routing and Switching	HUAWEI, CISCO, JUNIPER, EXTREME.
Telecommunications Networks	U2000, PRTG, STRM, DeviceExpert.

Licenses & certifications

Oct. 2020	Credential ID: 5EKD6ZY2PW72, Deep Learning Specialization	deeplearning.ai
Oct. 2020	Credential ID: EFMBB5WTQSYN, Neural Networks and Deep Learning	deeplearning.ai
Oct. 2020	Credential ID: FG8N2NLTR5FP, Convolutional Neural Networks	deeplearning.ai
Oct. 2020	Credential ID: 473XTBFS4RKV, Sequence Models	deeplearning.ai
Oct. 2020	Credential ID: EJ33G2B74FX8, Structuring Machine Learning Projects	deeplearning.ai
Oct. 2020	Credential ID: HP9GL7595JM6, Improving DNN: Hyperparameter tuning, Regularization and Optim[...]	deeplearning.ai

Scientific and Networking Programs (selection)

Apr. 2024	Talk, Scientific Machine Learning	Stockholm, SWE
Apr. 2024	Talk, From Theory to Practice Workshop	Kigali, RWD
Oct. 2023	Talk, CRUNCH group 'home of PINNs and DeepONet' seminar series	Rhode Island, USA
Sep. 2023	Talk, Kaiserslautern Applied and Industrial Mathematics Days	Kaiserslautern, GER
Sep. 2023	Talk, Applied Inverse Conference	Göttingen, GER
Jul. 2023	Talk & Poster, Maths for Deep Learning: Deep Learning for Computational Physics Conference	London, UK
Dec. 2022	Talk, African Institute for Mathematical Sciences, Cameroon Seminar	Limbe, CMR
Nov. 2022	Talk, Deep Learning IndabaX	Kindu, RDC
Sep. 2022	Talk, Machine Learning and Engineering Days	Hamburg, GER
Jul. 2022	Talk, World Congress on Computational Mechanics	Tokyo, JPN
Jan. 2016	Member, Organizing Committee, 1st Freshmen and Final Year Students' Conference	Yaounde, CMR
Apr. 2014	Organizer & Co-director, 3rd Engineering and Entrepreneurship Conference	Yaounde, CMR

Extracurricular and Volunteer Activities

HILPharma (Drug and Medical Awareness NGO)

Buea, Cameroon

TECHNICAL ADVISER/CONSULTANT

Jun. 2017 - Dec. 2019

- Gained experience in Social Media Marketing and Outreach.
- Reached out to 10000 youths on the issue of Drug Abuse in November 2018.

WCPN (World Changers' Professional Network)

Yaounde, Cameroon

FINANCE MANAGER AND ADMINISTRATOR

Sep. 2013 - Dec. 2019

- Gained expertise in NGO (Non-Governmental Organization) Management and Finance Administration.
- Provided Mentorship platform for University Students.
- Held several programs and outreaches to inspire and motivate youths in academics.

ASPY (Association of Anglophone Students of Polytechnique Yaounde)

Yaounde, Cameroon

PUBLIC RELATIONS OFFICER AND VICE PRESIDENT

Sep. 2010 - Jul. 2015

- Liaison between potential partners and alumni with students body with
- Provided mentorship platform for Engineering students.
- Held and organized programs and outreaches to inspire and motivate pre-engineering and engineering students.

References

Prof. Dr. Dr. hc Peter Maaß
ZeTEM, UNIVERSITY OF BREMEN
✉ pmaass@uni-bremen.de

Prof. Dr. Georg Bader
AIMS, GHANA & BRANDENBURGISCHE TECHNISCHE UNIVERSITÄT
✉ georg.bader@b-tu.de, georg@aims.edu.gh